



Confederation of Indian Industry



Centre of Excellence
for Competitiveness for SMEs



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Case Study Presentation By

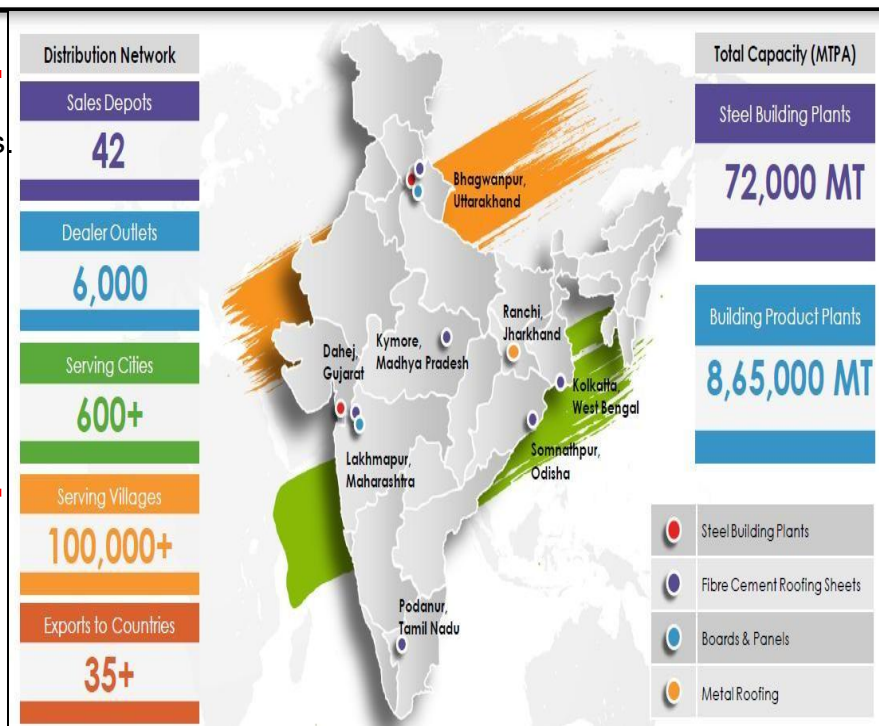
Everest Industries Limited
(Podanur Works, Coimbatore)

Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

Our Business:

- **Building Products (63%)** – include fiber cement roofing sheets, fiber cement boards, solid wall panels.
- **Steel buildings (37%)** – offers customized building solutions like Pre-Engineered Steel buildings, metal roofing sheets and cladding.



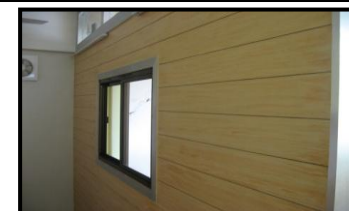
Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Project Title: Reducing power consumption by eliminating vacuum pump & 60 Hp motor.

Project Start Date: 15/03/2018 & **End Date:** 05/04/2018

Implemented Area: Sheeting Machine -1 Shop floor, Podanur Works, Coimbatore

Cross Functional Team:



Mr. M. Palavesa Moorthy
Prod. Shift Incharge



Mr. M. Madan Raj
Prod. Shift Incharge



Team Leader
Mr. Y. Kesavaraj
SM1 Line Incharge



Mr. K. Senthil Kumar
Electrical Supervisor



Mr. N. Anil Kumar
QS & BE



Mr. Sivadasan
Machine Driver



Mr. Madhialagan
Machine Driver



Mr. Peter
Machine Driver

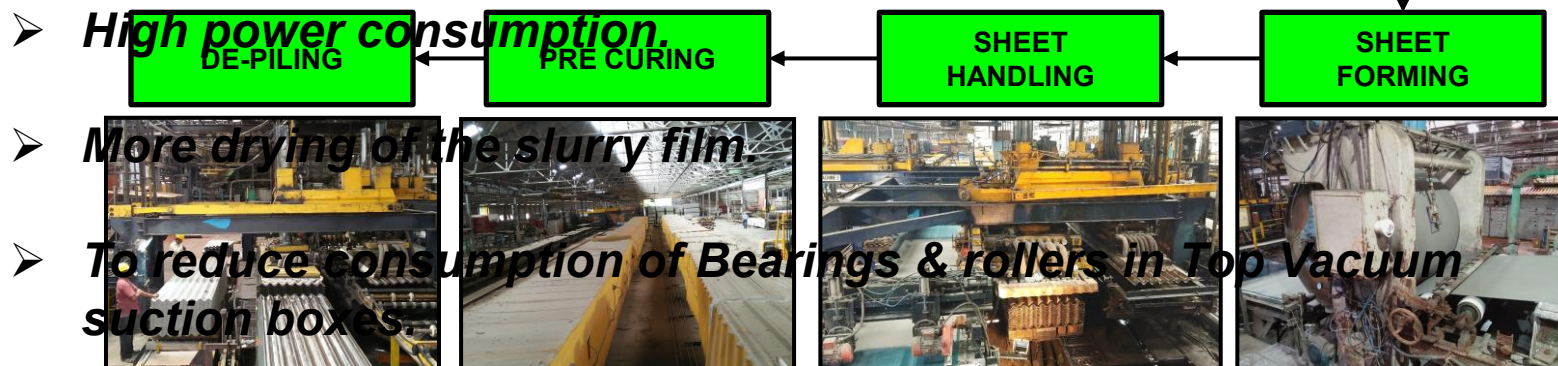
Process Flow:

Reducing Power consumption by eliminating 1 vacuum pump and 60Hp motor.

(Target:- To use 2 vacuum Pumps instead of 3 vacuum pumps)



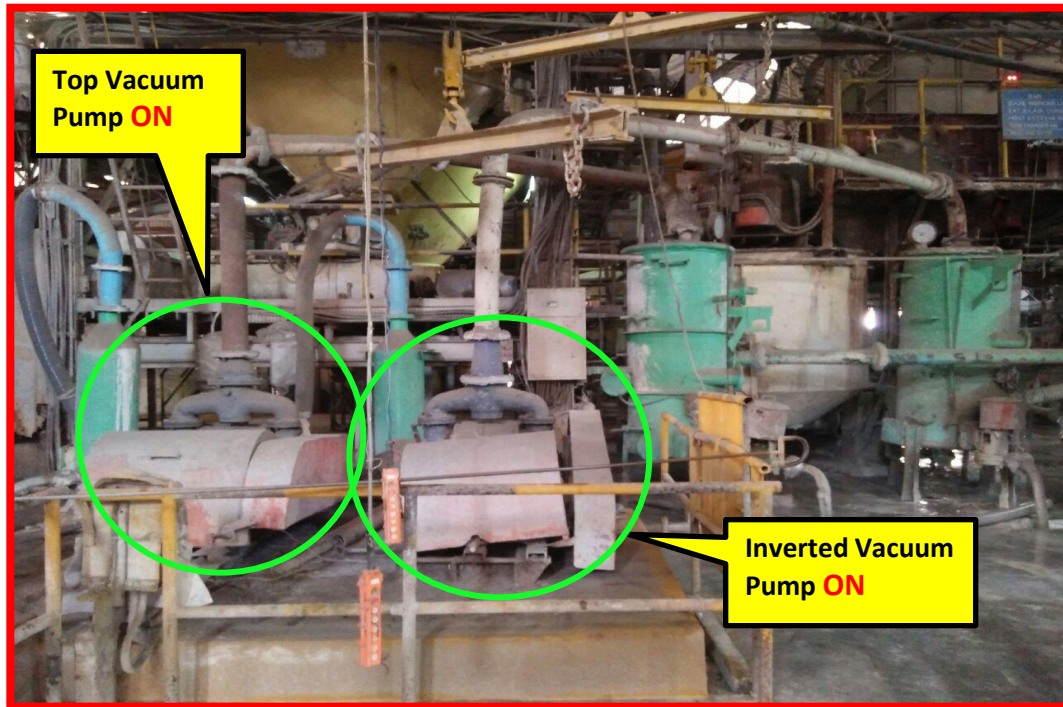
Reason for selecting THEME :



- **High power consumption.**
- **More drying of the slurry film.**
- **To reduce consumption of Bearings & rollers in Top Vacuum suction boxes.**

Problem :

More power consumption by vacuum pumps due to high vacuum use. If we reduce Vacuum below the optimum requirement we are facing problems like Layer falling from felt, Layer transferring problem to felt & over concentration of slurry in Vat.



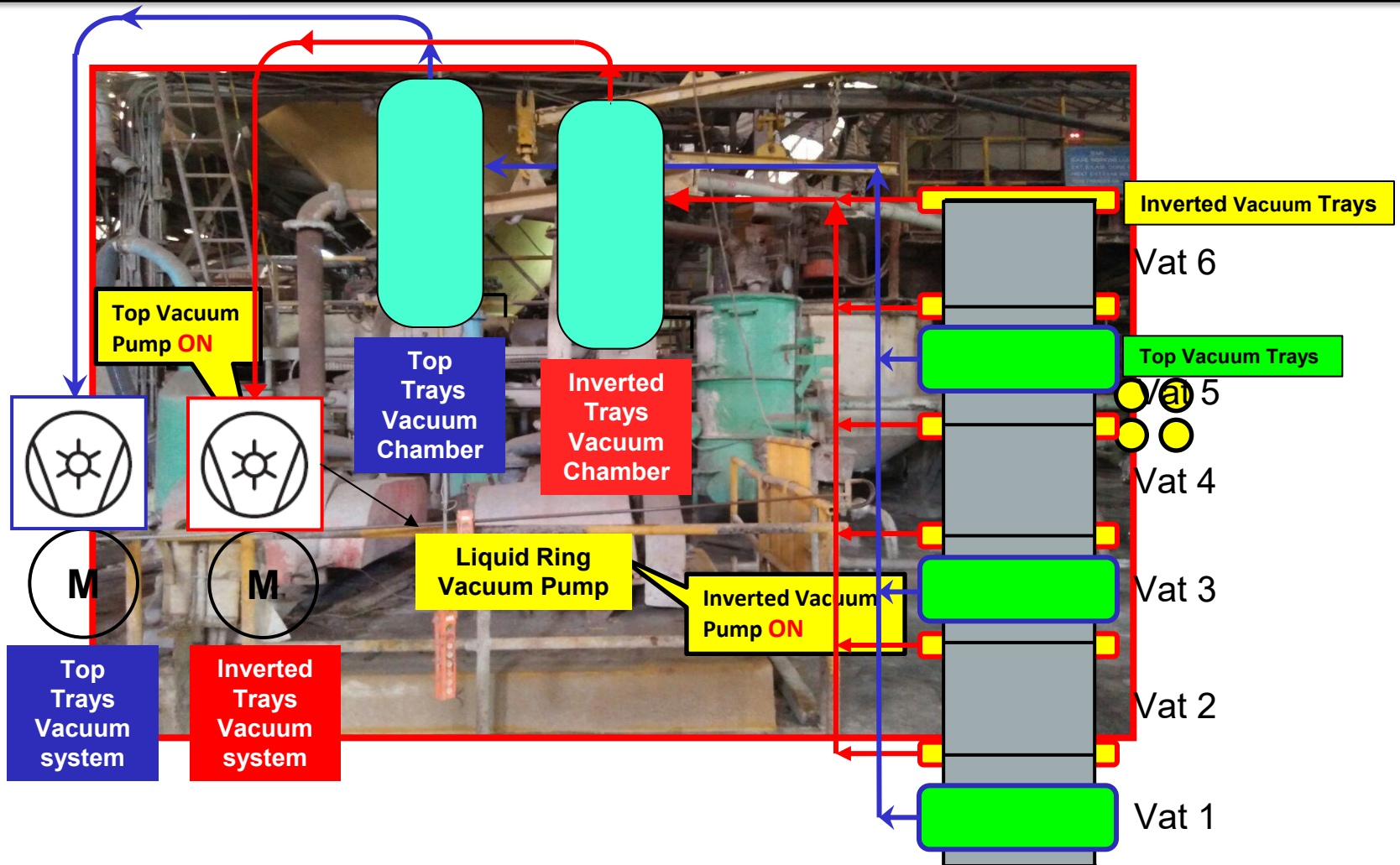
INVERTED VACUUM TRAY



TOP VACUUM TRAY

- Inverted suction Trays opening is high. So, to maintain required vacuum intensity, 1 dedicated vacuum pump is used for 6 inverted suction trays.
- Due to use of Rollers with bearing arrangement for decreasing the load on the main drive there is less opening area for Top suction Trays. To compensate the vacuum requirement for drying the film we are using 3 Top suction trays and using 1 dedicated vacuum pump.

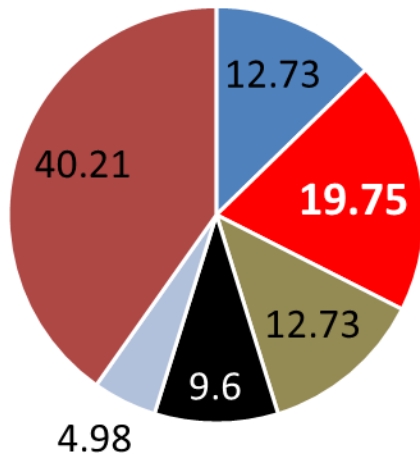
EVEREST Problem Description: Before Situation Process



- In the picture, we can see the flow of Vacuum through the pipe lines & vacuum chambers towards Top & Inverted vacuum pumps when both the vacuum pumps are in **ON** Condition.

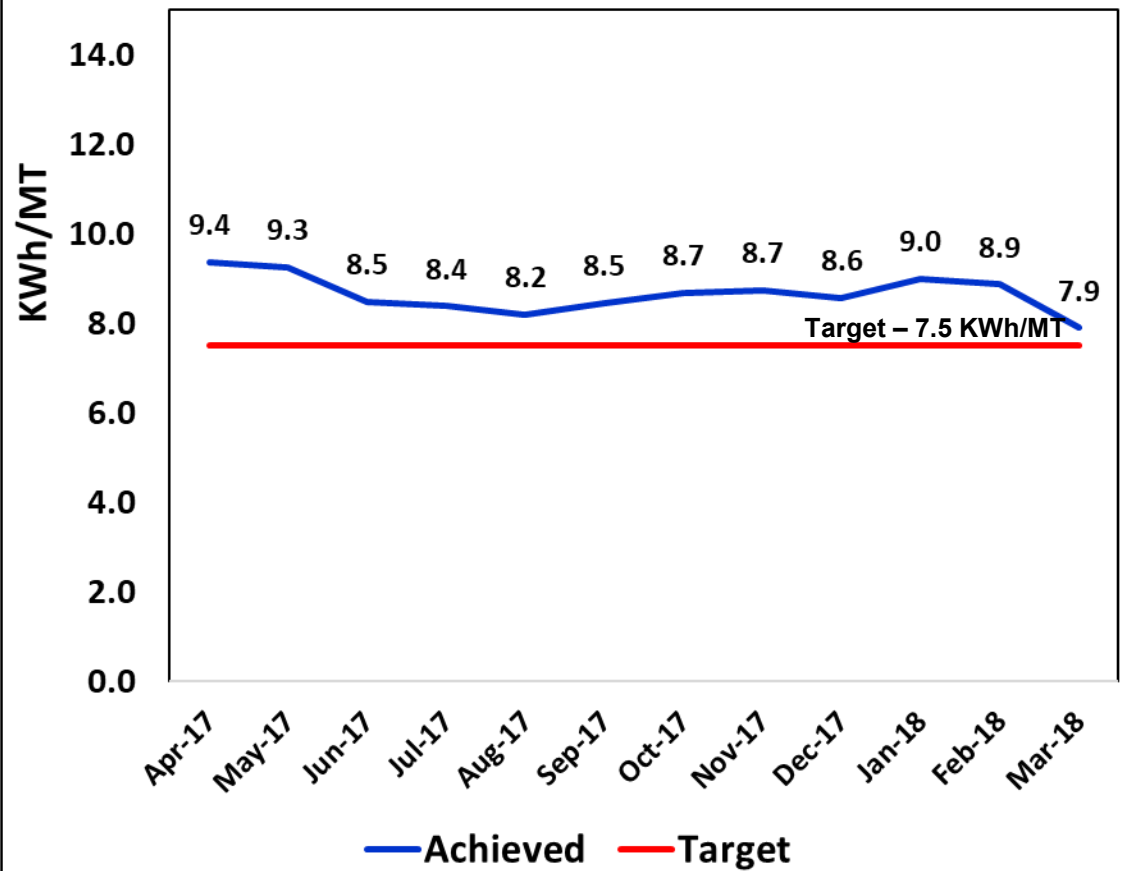
- Previous Monthly Trend of the Power consumption by Vacuum Pumps.

Power consumption(%)

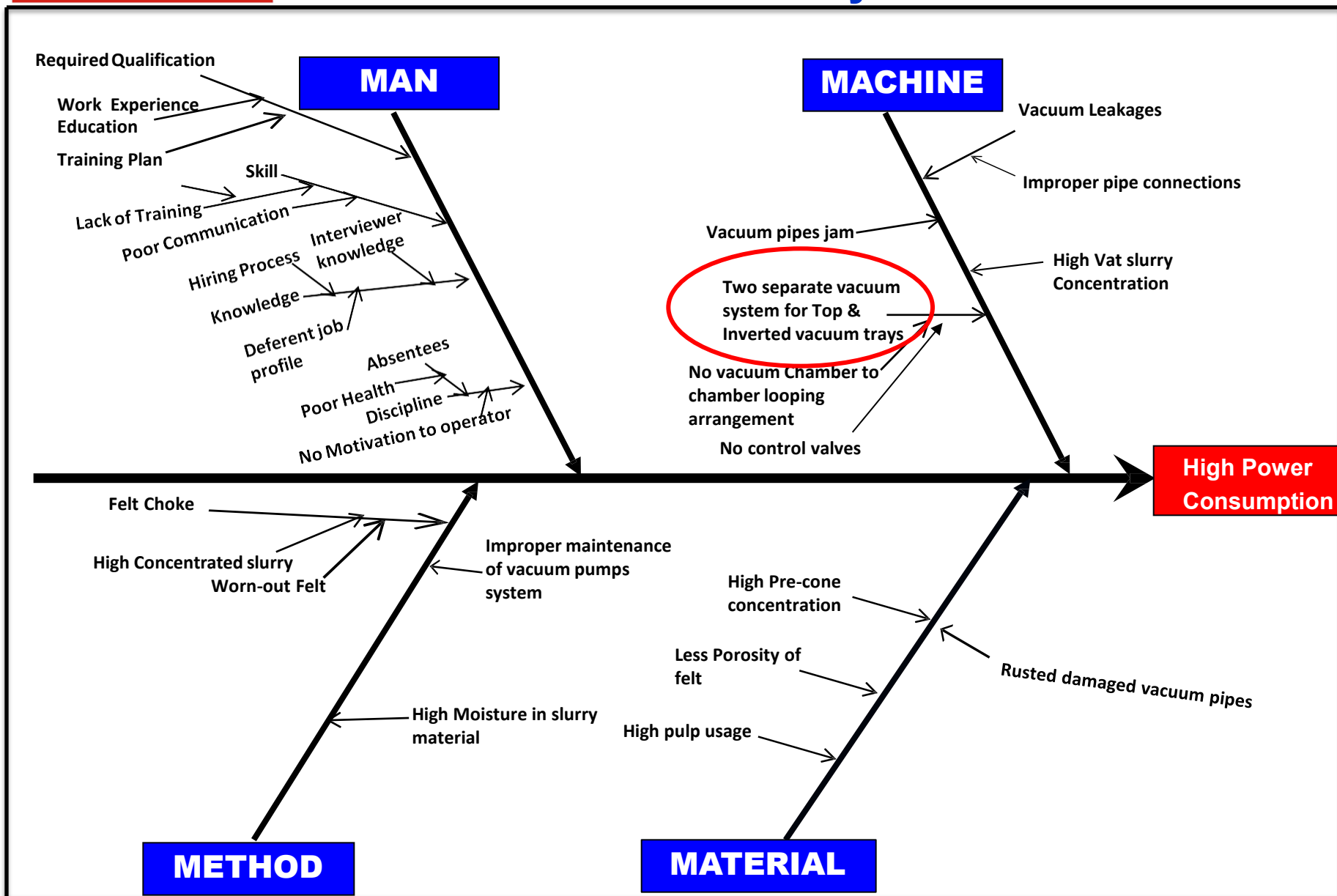


- Maindrive motor
- Vacuum pump motor
- Hydropulper motor
- Backwater pump motor
- Mixer's Stirrer motor
- Others

PW1(Vacuum Pumps)



S.NO	CONTAINMENT ACTIONS	RESPONSIBILITY	TARGET DATE	STATUS
1.	Reducing the Vacuum intensity	Prod. Shift In-charges / Line In-charge	15/03/2018	Completed
2.	Collecting data & ideas for reducing the power consumption	Shift In-charges / Line In-charge / Electrical Dept	15/03/2018	Completed



All the causes are validated for significant and non significant

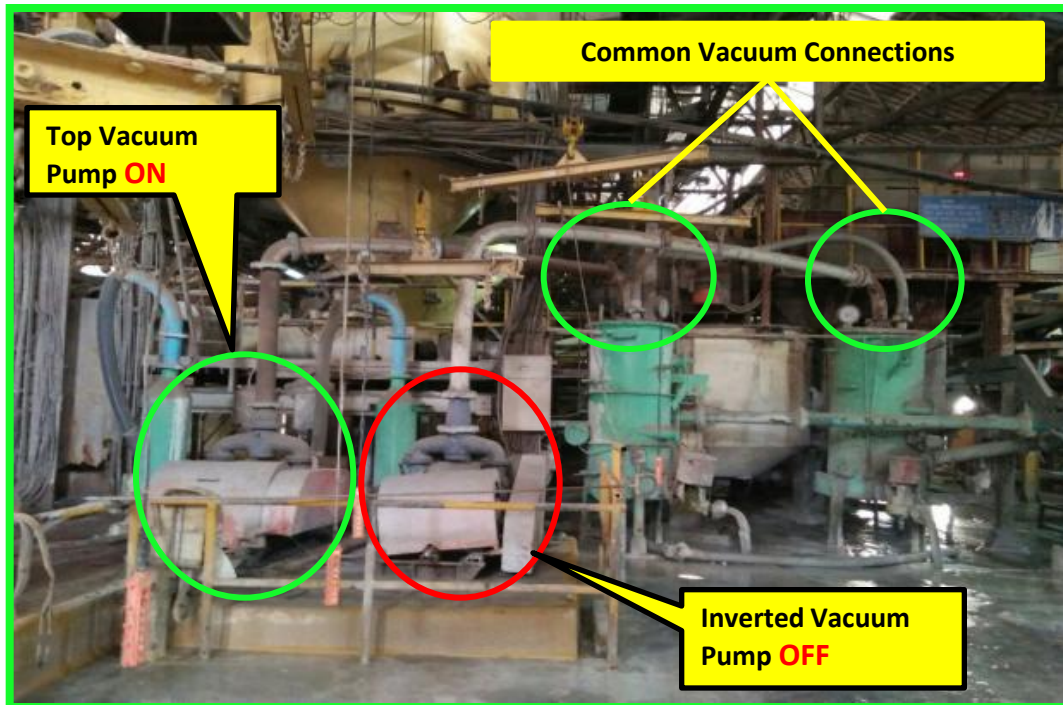
4M	Potential Causes	Observation	Confirmation
MAN	Required Qualification	skilled operator deputed on the machine.	Non Significant
	Skill	Aware machine operator deputed on the machine.	Non Significant
	Knowledge	Knowledgeable Operator deputed on the machine.	Non Significant
	Discipline	Operator has good discipline towards work.	Non Significant
MACHINE	Vacuum leakages	No Vacuum leakages in vacuum pipe connections.	Non Significant
	Vacuum pipes jam	No Slurry material setting in vacuum pipe lines.	Non Significant
	Two separate vacuum system for Top & Inverted vacuum trays.	There is No common connections b/w vacuum systems.	Significant
	High Vat Slurry concentration	High vat slurry concentration is maintain properly and regular vat consistency reading are taken.	Non Significant

All the causes are validated for significant and non significant

4M	Potential Causes	Observation	Confirmation
METHOD	Felt Choke	Proper cleaning of felt is done all the time.	Non Significant
	Improper maintenance of vacuum pumps.	Maintenance of vacuum pumps is done frequently as per schedule.	Non Significant
	High moisture in slurry material	Dilution of slurry with water is maintained perfectly.	Non Significant
MATERIAL	Rusted damaged vacuum pipes	Pipes are in good condition.	Non Significant
	High Pre-cone concentration	Pre-cone concentration filtration is within the specification limit.	Non Significant
	Less felt porosity	Quality of felt is according to the required specification. Quality is checked during inspection before approving felt.	Non Significant
	High pulp usage	Usage of pulp is within the specifies limit.	Non Significant

Problem Statement	More power consumption by vacuum pumps
1st why?	Why Power consumption by vacuum pumps is more?
Answer:	Using more vacuum
2nd why?	Why more vacuum is being used?
Answer:	Two Separate vacuum pumps are used for Inverted & Top suction boxes.
3rd why	Why Two Separate vacuum pumps are used for Inverted & Top suction boxes?
Answer:	Opening area of the inverted vacuum trays are high. To maintain required vacuum intensity separate vacuum pump is used for inverted vacuum suction boxes. And for Top suction boxes 1 vacuum pump is used.
Root Cause:	Design constraint of the Vacuum suction System.
Solution	1. To reduce the opening area of Inverted vacuum suction boxes.
	2. To provide looping arrangement between Top & Inverted vacuum chambers.

S. No	Actions	Resp.	Start Date	Target Date	Status
1	Modify the opening area of Bottom vacuum suction boxes by decreasing the opening area & volume of suction boxes.	Line In-charge	15/03/2018	30/03/18	Completed
2	Increase the opening area of the Top vacuum suction boxes by removing the rollers & bearings in the suction boxes and install Poly-picks.	Line In-charge	15/03/2018	30/03/18	Completed
3	Remove one Top vacuum suction box and to use only Two top suction boxes.	Line In-charge	20/03/2018	30/03/18	Completed
4	Provide looping arrangement b/w Top & Inverted vacuum chambers	Line In-charge	26/03/2018	02/04/18	Completed



INVERTED VACUUM TRAY



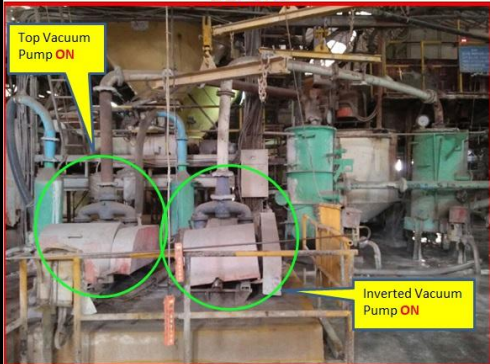
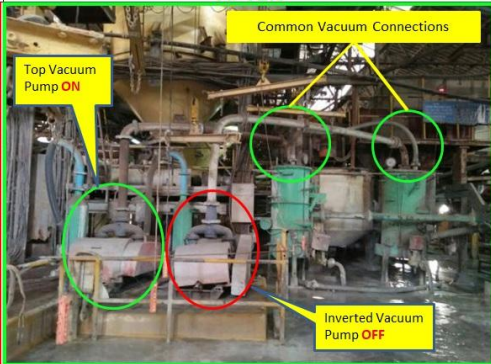
TOP VACUUM TRAY

- Modified vacuum suction boxes by decreasing opening area of the 6 inverted vacuum trays and removed rollers from 2 Top Vacuum suction boxes. By this we are now using only 1 vacuum pump for 2 Top & 6 Inverted suction boxes instead of using Two vacuum pumps.
- Modified the vacuum circuit by providing common vacuum connections b/w Top & Inverted vacuum chambers.
- By this we have stopped using 1 Vacuum pump, 60 Hp motor, Rollers & Bearings in vacuum trays & other consumables for maintenance.

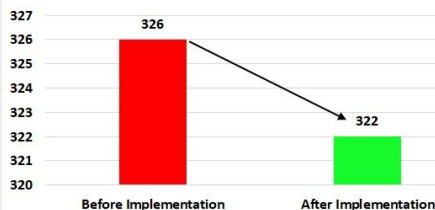


- In the above picture, Top vacuum pump is **ON** bottom vacuum pump is **OFF**.
- Vacuum from Top vacuum trays flows towards Top Vacuum Pump through vacuum chamber & pipe lines.
- Vacuum from Inverted vacuum trays flows towards Top vacuum pump through pipe lines, then to Inverted vacuum chamber and then to Top vacuum chamber, avoiding the flow towards Inverted vacuum pump.

- Implementation of Permanent corrective action as kaizen.

KAIZEN SHEET	Department	Production	Works	BW	SW	CW	KW	LW	PW	NW	BW- ESBS	EVEREST		
Kaizen Number	Line / Section	SM1	Result	P			Q			C	D		S	M
PW/KZN/04/2018	Machine /Equipment	Vacuum Pump		Productivity			Quality			Cost	Delivery		Safety	Morale
KAIZEN DESCRIPTION : Usage of only one vacuum pump for Top & Inverted vacuum trays instead of two vacuum pumps.		IDEA / Thought : To use only one vacuum pump for Top & Inverted vacuum trays.												
Problem / Present Status		Counter Measure						Target						
Previously for Top & Inverted vacuum separate vacuum pumps are used. (60 Hp + 60 Hp) = 120 Hp.		By connecting common vacuum line from chamber to chamber with looping arrangement we switched off 1 vacuum pump of 60Hp and using only one vacuum pump for Top & Inverted vacuum trays.						Kaizen Start Date Tuesday, April 3, 2018						
								Kaizen Finish Date Tuesday, April 3, 2018						
BEFORE		AFTER						Team Leader Y. Kesavaraj (SM1 Line Incharge)						
								Team Members : M. Palavesa Moothy (Shift officer), M. Madan Raj (Shift Officer), Sivadasan (Machine Driver), Peter (Machine Driver), Madhialagan (Machine Driver), K. Senthil Kumar (Electrical Supervisor)						
								Investment cost : Rs. 5000/-						
								Benefits : Motor Drive						
								Power saving = $\sqrt{3} \times V \times I \times \cos\phi$						
								= $(\sqrt{3} \times 415 \times 70 \times 0.8) / 1000$						
								= $40.25 \times 24 \text{ hrs} \times 28 \text{ Production days}$						
								= 27049 Units/Month						
								Power saving = $\sqrt{3} \times V \times I \times \cos\phi$						
								= $(\sqrt{3} \times 415 \times 5 \times 0.8) / 1000$						
								= 2.875 x 24 hrs x 28 Prod. Days						
								= 1932 Units/Month						
								Power saving per Month = (27049 + 1932) Units x 8 Rs. Per Unit = Rs. 2,31,848/-						
								Power saving per Year = Rs. 27,82,176/-						
								KAIZEN SUSTAINANCE :						
Why Why Analysis		GRAPHS						WHAT TO DO						
Problem :- Power consumption of vacuum pumps is more.								Regular Inspection and Maintenance.						
Why 1. From previously we are using separate vacuum pumps of 60Hp each for both the top & inverted vacuum trays.								HOW TO DO						
Why 2. There is no common vacuum line from chamber to chamber with looping arrangement.								Training and Instructions need to given to the Process persons.						
Why 3.								FREQUENCY						
Why 4.								Weekly						
Why 5.								Scope Of Horizontal Deployment						
Action Taken : By connecting common vacuum line from vacuum chamber to chamber we switched off 1 vacuum pump of 60Hp.								BW-ESBS						
								NW						
								BW						
								CW						
								KW						
								LW						
								PW						
								SW						
								Equipment						
								Target Date						
								Resp						
								Status						

Power consumption in Units per 1000 M²N



Calculation of Power consumption & Cost Savings of Vacuum Pumps:

- Previously we are using 2 Vacuum Pumps motor of 60Hp & 1 Vacuum Pump of 40Hp.
- Now we are using 1 Vacuum Pump motor of 60Hp & 1 Vacuum Pump motor of 40Hp.

$$\text{Power Saving} = \sqrt{3} \times V \times I \times \cos\Phi$$

$$= (\sqrt{3} \times 415 \times 70 \times 0.8) / 1000 = 40.25 \text{ kW}$$

$$\text{Monthly Savings} = 40.25 \times 24 \text{ Hrs} \times 28 \text{ Production days}$$

$$= 27,049 \text{ kWh/Month By eliminating 60Hp Vacuum Pump motor}$$

$$\text{Cost Saving per Month} = 27,049 \times 8 \text{ Rs. Per Unit}$$

$$= \text{Rs. 2,16,392/- per month}$$

$$\text{Cost Saving per Year} = 25.96 \text{ Lakh/Annum}$$

- Cost Saving by eliminating Roller suction trays & Bearings = 1.72 Lakh/Annum
- Cost Saving by eliminating Maintenance of 1 Vacuum Pump = Rs. 0.17 Lakh/Annum
- **Total Cost Saving is 27.95 Lakh/Annum**

Before Situation:

Vacuum suction boxes opening Area

Area of 1 Suction box = 2044 CM²

Area of 1 Suction box = 1898 CM²

Inverted Vacuum Trays

Top Vacuum Trays

After Situation:

Area of 1 suction box = 1168 CM²

Area of 1 Suction box = 2920 CM²

Inverted Vacuum Trays

Top Vacuum Trays

Tangible Benefits

- Average Monthly power consumption of vacuum pumps is reduced from 65745 Units to 45276 Units.
- 31.13% reduction in Vacuum pumps power consumption.
- Use of rollers and bearings in Top vacuum trays is eliminated. And maintenance of them is eliminated.
- Eliminated the use of 1 vacuum pump and motor of 60Hp.
- Potential cost saving of 27.95 Lakh./Annum.
- Problems like slurry layer falling from felt, layer transfer problem to felt & Slurry over concentration in Vats has reduced.

In-tangible Benefits

- Morale increased due to reduction in power consumption.
- Morale increased due to reduction in effort for maintenance of vacuum pumps, rollers & bearings of Top vacuum trays.

- Training on Standard Operating Procedures are given to all the maintenance people for regular inspection & maintenance.

	EVEREST INDUSTRIES LIMITED PODANUR WORKS IS/ISO 9001 : 2015 STANDARD OPERATING PROCEDURE	ISSUE NO : 01 REV NO : 01 PAGE : 1 OF 1 DATE : 01/02/2018
	Standard Operating Procedure: Starting of Vacuum Pump 1. During machine starting start 1 vacuum pump. 2. Ensure chamber to chamber looping line valve should be in open condition. 3. Check the Vacuum Gauges in both the vacuum chambers for verifying if the vacuum increased or not. 4. Maintain the Vacuum as per our standard requirement. 5. Repeat the process during every machine startup.	

Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Horizontal Deployment:

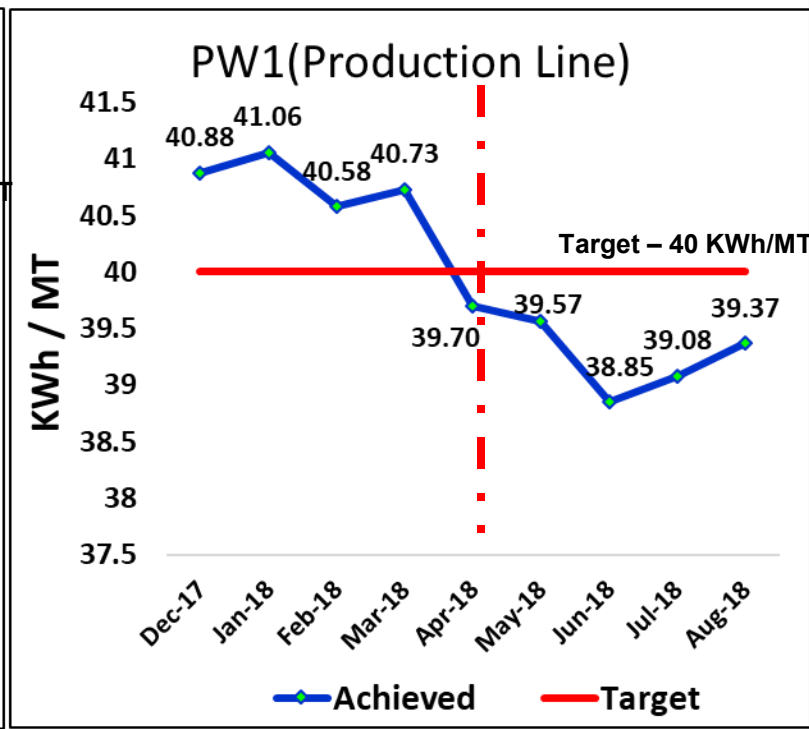
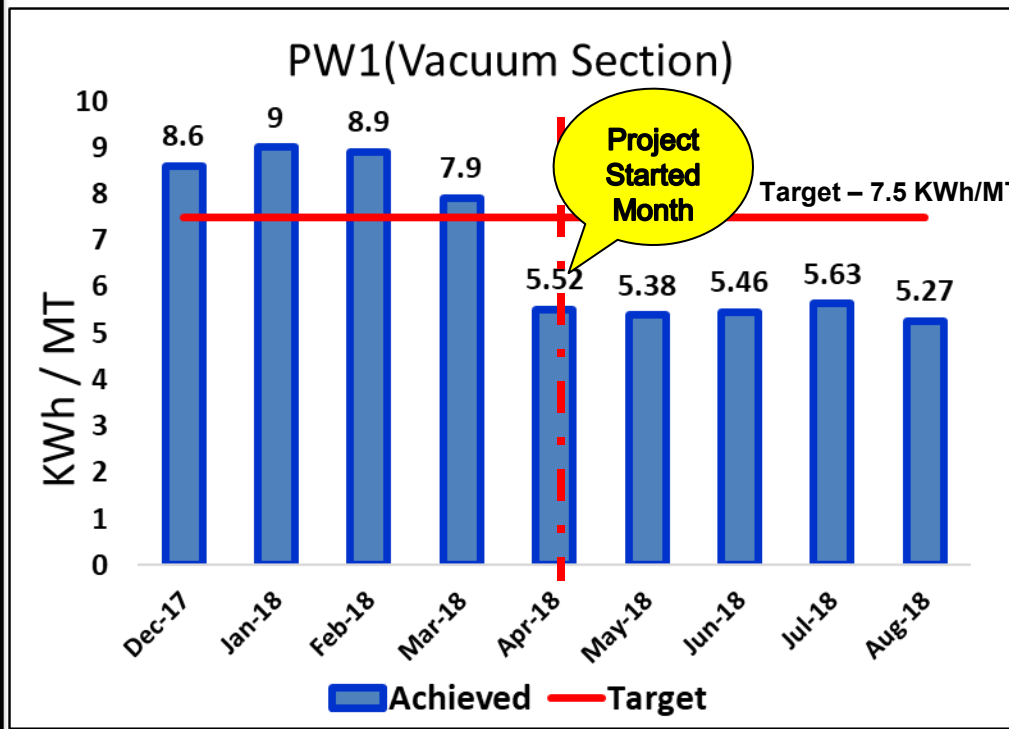
Factory	Target Date	Status
1.Bhagwanpur Factory, Roorkee	31.12.2018	Under Implementation
2.Kolkata Factory, Kolkata	31.01.2019	Under Implementation
3.Kymore Factory, Madhya Pradesh	15.06.2018	Implemented
4.Podanur Factory, Coimbatore	30.04.2018	Implemented in both the lines
5.Somnathpur Factory, Odisha	30.09.2018	Implemented

Objective:

Reducing Power consumption by eliminating 1 vacuum pump and 60Hp motor.

(Target:- To use 2 vacuum Pumps instead of 3)

- Monthly Trend of reduction of power consumption by Vacuum Pumps After Implementing the project.



- By implementing this project there is significant decrease in the use of Power consumption & spares.
- Problem solving is an effective tool which is used for knowing the hidden facts of more power consumption. Further this process can be used in other area for solving the problems.
- Gaining awareness on how the vacuum system works.

- Rewards and recognition to entire team members



Thank You